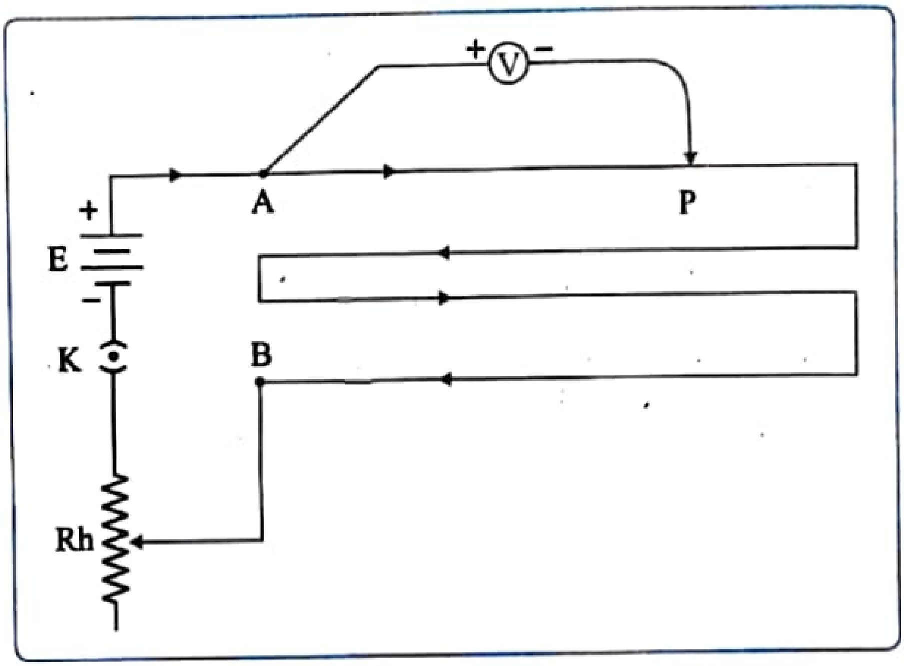
ACTIVITY NO. 7
VARIATION OF POTENTIAL DROP WITH LENGTH OF WIRE

Aim: To study the variation in potential drop with length of a wire for a steady current.

Apparatus : Potentiometer, battery (or driving cell), plug key, voltmeter, rheostat, jockey, connecting wires.

Circuit diagram :



Formula : For a potentiometer wire of uniform density and cross-sectional area carrying a steady current, potential drop (V) is proportional to length (ℓ) of the wire.
Potential gradient $k = V/\ell$

Procedure :

1. Arrange and connect the circuit as shown in the circuit diagram Close the key K.
2. Touch the jockey at length 60 cm of the wire from end A. Note down the voltmeter reading.
3. Touch the jockey at different length such as, 120 cm, 180cm, 240cm of the wire from end A and note the voltmeter reading in each case.

Observations :

Least count of voltmeter = 0.5 V

Ob. No.	Length of potentiometer wire 'ℓ' cm	Voltmeter Reading 'V' volt	Ratio $\frac{V}{\ell}$ V/cm
1	50	0.15	0.003
2	100	0.35	0.003
3	150	0.55	0.003

Calculations :

$$\frac{r}{l} = \frac{0.15}{50} = 0.003$$

$$\frac{r}{l} = \frac{0.35}{100} = 0.003$$

$$\frac{r}{l} = \frac{0.55}{150} = 0.003$$

Result :

Potential gradient V/L= 0.003 = constant,

Precautions :

1. All the connections should be tight.
2. By closing the key 'K' adjust the rheostat so that the voltmeter gives full scale deflection.

Questions

1. Define potential gradient.

Potential gradient is the rate of change of potential with respect to displacement.

. What is advantage of potentiometer over voltmeter?

A potentiometer is more accurate than a voltmeter because it measures potential difference without drawing any current from the circuit. Therefore, there is no loading effect and the measurement is more precise.

Remark and sign of teacher :